MILITARY SPECIFICATION

RATE INTEGRATING GYRO
MS 3391

This amendment forms a part of Military Specification MIL-R-23750A(AS) dated 1 August 1968 and is approved for use by the Naval Air System Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

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Add the following new paragraph:

"3.5.3.7.3 Gimbal angular limits - The gimbal angular limit is expressed as the maximum angular displacement Of the gimbal about its output axis with respect to its support. The allowable limit is expressed as:

$$3.33\frac{H}{B} > \theta_G \text{ limit}$$

where: H = angular momentum (dyn-cm-sec)
B = coefficient of damping (dyne-cm-sec)
θ_G limit =angular limit of gimbal (deg)

When tested in accordance with 4.8.1.6.3 the pickoff voltage shall reach the limiting voltage when the gyro is rotated through 3.3 ± 0.1 angular degrees."

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Add the following new paragraph:

"4.8.1.6.3 Gimbal angular limits - Orient the gyro as in 4.8.1.2.1. Close the normal gyro loop to null the gyro gimbal position. Disconnect pins J and T and rotate the gyro through a CW angle large enough to cause the gimbal to be driven into the gimbal limit stops. Record the pickoff signal. Repeat the test with a CCW angular rotation. Care must be exercised to perform the tests rapidly enough to prevent drift away from null after disconnecting pins J and T prior to applying the input axis motion. Repeat the tests except rotate the gyro to the angles specified in 3.5.3.7.3. The pickoff voltage shall reach the maximum limiting voltages recorded shove. Care must be exercised to perform the final readings rapidly enough to prevent drift away from the final angular gimbal position."

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